

Supplementary Material

for

Determination of activities and condensation temperatures of $\text{GaO}_{1.5}$ and $\text{InO}_{1.5}$ in anorthite-diopside eutectic melts by Knudsen Effusion Mass Spectrometry

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Partial pressures above anorthite-diopside eutectic melts by Knudsen effusion mass spectrometry

1_low (KN_07-11-19_Ga-In-Tl_1000)

Table S1														
Polytherm														
T (K)	Ga (Pa)	Ga _{corr} (Pa)	ln (Pa)	ln _{corr} (Pa)	O _{2, calc} (Pa)	K (Ga)	SD	γ(GaO1.5)	SD	K (ln)	SD	γ(InO1.5)	SD	AIW
1586.4	4.04E-05	4.04E-05	6.69E-05	6.69E-05	0.000112	4.54E-15	2.28E-15	6.32E-02	5.58E-02	1.87E-12	1.08E-12	5.27E-02	4.37E-02	1.66
1598.0	4.37E-05	4.37E-05	7.25E-05	7.25E-05	0.000137	5.73E-15	2.81E-15	4.78E-02	4.18E-02	2.36E-12	1.33E-12	4.40E-02	3.61E-02	1.62
1608.6	4.49E-05	4.49E-05	9.76E-05	9.76E-05	0.000174	7.02E-15	3.40E-15	3.69E-02	3.22E-02	3.79E-12	2.12E-12	4.86E-02	3.97E-02	1.61
1619.2	5.16E-05	5.16E-05	8.90E-05	8.90E-05	0.000206	9.17E-15	4.30E-15	3.05E-02	2.64E-02	3.93E-12	2.14E-12	3.49E-02	2.81E-02	1.57
1630.8	6.45E-05	6.45E-05	1.17E-04	1.17E-04	0.00027	1.40E-14	6.56E-15	2.85E-02	2.46E-02	6.33E-12	3.43E-12	3.77E-02	3.03E-02	1.57
1642.5	1.10E-04	1.10E-04	1.23E-04	1.23E-04	0.000355	2.93E-14	1.37E-14	3.67E-02	3.17E-02	8.19E-12	4.44E-12	3.29E-02	2.65E-02	1.57
1653.0	9.98E-05	9.98E-05	1.46E-04	1.46E-04	0.000429	3.08E-14	1.40E-14	2.48E-02	2.13E-02	1.12E-11	5.94E-12	3.15E-02	2.52E-02	1.54
1664.7	1.45E-04	1.45E-04	1.60E-04	1.60E-04	0.000555	5.40E-14	2.46E-14	2.72E-02	2.33E-02	1.49E-11	7.88E-12	2.86E-02	2.28E-02	1.54
1674.2	1.66E-04	1.66E-04	1.97E-04	1.97E-04	0.000681	7.24E-14	3.27E-14	2.48E-02	2.12E-02	2.14E-11	1.12E-11	3.01E-02	2.39E-02	1.54
1684.8	2.07E-04	2.07E-04	2.07E-04	2.07E-04	0.000844	1.06E-13	4.75E-14	2.39E-02	2.04E-02	2.64E-11	1.38E-11	2.65E-02	2.10E-02	1.53
1695.4	2.73E-04	2.73E-04	2.34E-04	2.34E-04	0.00106	1.66E-13	7.39E-14	2.46E-02	2.10E-02	3.54E-11	1.84E-11	2.54E-02	2.01E-02	1.53

Table S2															
Isotherm															
T (K)	Ga (Pa)	Ga ^{corr} (Pa)	ln (Pa)	ln ^{corr} (Pa)	O _{2, calc} (Pa)	K (Ga)	SD	γ(GaO1.5)	SD	K (ln)	SD	γ(lnO1.5)	SD	ΔIW	
1741.9	8.81E-04	8.81E-04	3.20E-04	3.20E-04	0.002761	1.39E-12	6.09E-13	3.53E-02	2.99E-02	1.23E-10	6.34E-11	2.11E-02	1.66E-02	1.51	
1741.9	9.46E-04	9.46E-04	3.47E-04	3.47E-04	0.002805	1.51E-12	6.66E-13	3.84E-02	3.26E-02	1.34E-10	7.00E-11	2.31E-02	1.83E-02	1.52	
1741.9	1.04E-03	1.04E-03	2.94E-04	2.94E-04	0.002834	1.68E-12	7.45E-13	4.26E-02	3.63E-02	1.15E-10	6.01E-11	1.97E-02	1.57E-02	1.52	
1741.9	1.19E-03	1.19E-03	3.21E-04	3.21E-04	0.002919	1.96E-12	8.81E-13	4.97E-02	4.25E-02	1.28E-10	6.77E-11	2.20E-02	1.75E-02	1.54	
1741.9	1.26E-03	1.26E-03	2.82E-04	2.82E-04	0.002937	2.08E-12	9.36E-13	5.27E-02	4.50E-02	1.13E-10	5.99E-11	1.94E-02	1.55E-02	1.54	
1743.0	1.21E-03	1.21E-03	2.92E-04	2.92E-04	0.002915	1.98E-12	8.91E-13	4.84E-02	4.13E-02	1.16E-10	6.15E-11	1.94E-02	1.54E-02	1.53	
1743.0	1.29E-03	1.29E-03	2.81E-04	2.81E-04	0.002952	2.14E-12	9.66E-13	5.22E-02	4.46E-02	1.13E-10	6.00E-11	1.88E-02	1.50E-02	1.53	
1740.8	1.13E-03	1.13E-03	2.86E-04	2.86E-04	0.002877	1.85E-12	8.25E-13	4.88E-02	4.16E-02	1.13E-10	5.93E-11	2.00E-02	1.59E-02	1.54	
1740.8	1.16E-03	1.16E-03	2.65E-04	2.65E-04	0.002881	1.89E-12	8.44E-13	4.99E-02	4.25E-02	1.04E-10	5.51E-11	1.86E-02	1.48E-02	1.54	
1740.8	1.20E-03	1.20E-03	2.77E-04	2.77E-04	0.002905	1.96E-12	8.81E-13	5.18E-02	4.42E-02	1.10E-10	5.82E-11	1.96E-02	1.56E-02	1.54	
1740.8	1.21E-03	1.21E-03	2.37E-04	2.37E-04	0.002896	1.98E-12	8.88E-13	5.23E-02	4.46E-02	9.38E-11	4.96E-11	1.67E-02	1.33E-02	1.54	
1741.9	1.11E-03	1.11E-03	2.80E-04	2.80E-04	0.00286	1.79E-12	7.99E-13	4.55E-02	3.88E-02	1.10E-10	5.78E-11	1.89E-02	1.50E-02	1.53	
1740.8	1.09E-03	1.09E-03	2.44E-04	2.44E-04	0.002839	1.76E-12	7.83E-13	4.66E-02	3.96E-02	9.52E-11	4.99E-11	1.69E-02	1.34E-02	1.53	
1740.8	1.01E-03	1.01E-03	2.52E-04	2.52E-04	0.002801	1.62E-12	7.13E-13	4.27E-02	3.62E-02	9.74E-11	5.08E-11	1.73E-02	1.37E-02	1.53	
1740.8	1.04E-03	1.04E-03	2.42E-04	2.42E-04	0.002813	1.67E-12	7.37E-13	4.40E-02	3.74E-02	9.40E-11	4.90E-11	1.67E-02	1.32E-02	1.53	
1740.8	9.66E-04	9.66E-04	2.27E-04	2.27E-04	0.002768	1.53E-12	6.70E-13	4.03E-02	3.42E-02	8.70E-11	4.51E-11	1.54E-02	1.22E-02	1.52	

Table S3

Polytherm													
T (K)	Ga (Pa)	Ga _{corr} (Pa)	GaO (Pa)	Ga ₂ (Pa)	Ga ₂ O (Pa)	Ga ₂ O _{corr} (Pa)	In (Pa)	In _{corr} (Pa)	InO (Pa)	In ₂ O (Pa)	In ₂ O _{corr} (Pa)	O ₂ , meas (Pa)	O ₂ , calc (Pa)
1401.3			6.20E-06	1.79E-05	3.74E-06	2.17E-05	2.01E-04	2.01E-04	3.59E-06	4.78E-06	4.78E-06	3.59E-04	9.29E-05
1413.9			6.29E-06	5.95E-06	9.10E-06	1.50E-05	2.06E-04	2.06E-04				7.67E-04	8.99E-05
1425.6	4.98E-05	4.98E-05	3.09E-06	5.85E-06	2.51E-06	8.37E-06	2.79E-04	2.79E-04	4.30E-06	2.65E-06	2.65E-06	5.14E-04	1.42E-04
1437.2	7.02E-05	7.02E-05		7.57E-06	1.88E-06	9.45E-06	3.69E-04	3.69E-04	1.04E-06			2.53E-04	1.87E-04
1449.9	1.08E-04	1.08E-04		2.54E-06	7.66E-06	1.02E-05	4.96E-04	4.96E-04				4.22E-04	2.57E-04
1461.6	1.55E-04	1.55E-04		4.53E-06	5.76E-06	1.03E-05	6.19E-04	6.19E-04	9.40E-06	3.08E-06	3.08E-06	2.38E-06	3.33E-04
1474.2	2.13E-04	2.13E-04	8.69E-07	3.21E-06	1.48E-06	4.68E-06	7.75E-04	7.75E-04	1.57E-06			6.63E-05	4.21E-04
1486.9	2.71E-04	2.71E-04		1.03E-05	1.49E-05	2.52E-05	8.40E-04	8.40E-04	3.51E-06	3.44E-06	3.44E-06	7.68E-04	4.88E-04
1498.6	3.59E-04	3.59E-04	7.64E-06	4.10E-06	5.57E-06	9.68E-06	8.23E-04	8.23E-04	2.25E-06			4.01E-04	5.21E-04
1511.3	4.23E-04	4.23E-04		2.28E-06	1.80E-05	2.02E-05	8.31E-04	8.31E-04	3.37E-06	4.38E-06	4.38E-06	1.44E-04	5.64E-04
1522.9	4.93E-04	4.93E-04	1.63E-05	1.32E-06	5.88E-06	7.20E-06	6.03E-04	6.03E-04		5.97E-06	5.97E-06	5.97E-06	5.10E-04
1535.6	4.93E-04	4.93E-04		7.37E-06	4.64E-06	1.20E-05	5.50E-04	5.50E-04	8.05E-06	1.82E-06	1.82E-06	4.51E-04	4.93E-04
1548.3	4.74E-04	4.74E-04		1.56E-06		1.56E-06	5.80E-04	5.80E-04	4.41E-06	6.39E-06	6.39E-06	4.19E-05	4.99E-04
1557.8	3.08E-04	3.08E-04	6.84E-07		5.39E-06	5.39E-06	7.59E-04	7.59E-04				4.91E-04	4.91E-04
1571.6	1.05E-04	1.05E-04		2.09E-06	6.41E-06	8.50E-06	3.67E-04	3.67E-04	2.72E-06	3.58E-06	3.58E-06	2.49E-04	2.49E-04
1588.5	1.34E-04	1.34E-04	3.47E-06	2.62E-05	2.68E-07	2.64E-05	4.41E-04	4.41E-04	1.41E-05			1.63E-04	3.26E-04
1601.2	1.24E-04	1.24E-04		2.46E-06	6.99E-06	9.45E-06	3.91E-04	3.91E-04	1.28E-05			6.86E-04	3.17E-04
1613.9	1.36E-04	1.36E-04			1.56E-07	1.56E-07	3.86E-04	3.86E-04		3.79E-06	3.79E-06	4.84E-05	3.51E-04
1626.6	1.42E-04	1.42E-04		4.82E-06	2.39E-07	5.06E-06	3.62E-04	3.62E-04	6.08E-06			3.92E-04	3.92E-04
1638.2	1.65E-04	1.65E-04	3.22E-06	4.96E-06	1.51E-06	6.47E-06	3.25E-04	3.25E-04	9.31E-06	3.91E-06	3.91E-06	7.20E-04	4.46E-04
1650.9	1.98E-04	1.98E-04		5.66E-06	9.96E-06	1.56E-05	3.04E-04	3.04E-04				1.29E-04	5.34E-04
1662.6	2.43E-04	2.43E-04	4.57E-06	2.51E-05	1.57E-05	4.08E-05	2.57E-04	2.57E-04	1.69E-05	4.03E-06	4.03E-06		6.48E-04
1674.2	3.05E-04	3.05E-04		6.36E-06	2.54E-06	8.89E-06	2.42E-04	2.42E-04				1.60E-04	7.75E-04
1685.8	3.68E-04	3.68E-04		1.41E-05	3.54E-06	1.76E-05	2.30E-04	2.30E-04				2.58E-04	9.60E-04
1699.6	4.44E-04	4.44E-04	1.23E-06	1.87E-06	8.91E-06	1.08E-05	2.36E-04	2.36E-04	4.14E-06	4.22E-06	4.22E-06	2.35E-04	1.24E-03
1712.3	5.91E-04	5.91E-04		1.02E-05	6.32E-06	1.65E-05	2.20E-04	2.20E-04				2.54E-04	1.59E-03
1719.7	7.00E-04	7.00E-04					2.24E-04	2.24E-04	1.96E-05			1.03E-03	1.85E-03
1733.4	9.71E-04	9.71E-04			1.02E-05	1.02E-05	2.03E-04	2.03E-04	5.24E-06			1.02E-03	2.46E-03

2_low (KN_06_03_20_Ga_In_1000)

Table S3. – cont.

<i>Polytherm</i>									
T (K)	K (Ga)	SD	$\gamma(\text{GaO}_{1.5})$	SD	K (ln)	SD	$\gamma(\text{InO}_{1.5})$	SD	AIW
1401.3					5.54E-13	6.23E-13	2.96E+01	3.77E+01	3.86
1413.9					5.54E-13	6.29E-13	1.66E+01	2.12E+01	3.67
1425.6	6.71E-15	4.41E-15	2.70E+02	2.65E+02	1.06E-12	1.22E-12	1.88E+01	2.44E+01	3.71
1437.2	1.16E-14	7.66E-15	2.48E+02	2.43E+02	1.72E-12	1.99E-12	1.82E+01	2.37E+01	3.67
1449.9	2.27E-14	1.49E-14	2.44E+02	2.39E+02	2.93E-12	3.39E-12	1.79E+01	2.32E+01	3.65
1461.6	3.95E-14	2.60E-14	2.30E+02	2.25E+02	4.45E-12	5.13E-12	1.65E+01	2.14E+01	3.61
1474.2	6.48E-14	4.27E-14	1.95E+02	1.91E+02	6.64E-12	7.68E-12	1.44E+01	1.88E+01	3.55
1486.9	9.21E-14	6.04E-14	1.45E+02	1.42E+02	8.04E-12	9.27E-12	1.03E+01	1.34E+01	3.45
1498.6	1.28E-13	8.43E-14	1.12E+02	1.10E+02	8.27E-12	9.56E-12	6.63E+00	8.61E+00	3.34
1511.3	1.60E-13	1.05E-13	7.47E+01	7.30E+01	8.86E-12	1.02E-11	4.27E+00	5.54E+00	3.22
1522.9	1.73E-13	1.13E-13	4.58E+01	4.48E+01	5.96E-12	6.87E-12	1.82E+00	2.36E+00	3.03
1535.6	1.69E-13	1.10E-13	2.43E+01	2.37E+01	5.30E-12	6.09E-12	9.87E-01	1.28E+00	2.87
1548.3	1.64E-13	1.06E-13	1.30E+01	1.26E+01	5.64E-12	6.46E-12	6.48E-01	8.37E-01	2.73
1557.8	1.05E-13	6.74E-14	5.34E+00	5.18E+00	7.29E-12	8.31E-12	5.86E-01	7.53E-01	2.61
1571.6	2.14E-14	1.30E-14	5.81E-01	5.50E-01	2.12E-12	2.34E-12	1.02E-01	1.28E-01	2.17
1588.5	3.36E-14	2.01E-14	4.26E-01	4.00E-01	3.12E-12	3.41E-12	8.14E-02	1.02E-01	2.10
1601.2	3.04E-14	1.75E-14	2.20E-01	2.04E-01	2.70E-12	2.90E-12	4.50E-02	5.52E-02	1.95
1613.9	3.60E-14	2.01E-14	1.51E-01	1.38E-01	2.89E-12	3.05E-12	3.07E-02	3.73E-02	1.86
1626.6	4.10E-14	2.20E-14	9.96E-02	8.99E-02	2.94E-12	3.04E-12	2.02E-02	2.41E-02	1.77
1638.2	5.23E-14	2.71E-14	7.79E-02	6.95E-02	2.91E-12	2.95E-12	1.35E-02	1.58E-02	1.71
1650.9	7.21E-14	3.61E-14	6.34E-02	5.60E-02	3.11E-12	3.11E-12	9.41E-03	1.09E-02	1.66
1662.6	1.02E-13	4.98E-14	5.59E-02	4.89E-02	3.04E-12	2.99E-12	6.26E-03	7.21E-03	1.63
1674.2	1.47E-13	6.99E-14	5.02E-02	4.36E-02	3.28E-12	3.19E-12	4.62E-03	5.28E-03	1.59
1685.8	2.07E-13	9.70E-14	4.47E-02	3.86E-02	3.65E-12	3.52E-12	3.54E-03	4.01E-03	1.57
1699.6	3.03E-13	1.39E-13	3.82E-02	3.28E-02	4.53E-12	4.33E-12	2.85E-03	3.21E-03	1.55
1712.3	4.87E-13	2.21E-13	3.77E-02	3.22E-02	5.11E-12	4.86E-12	2.16E-03	2.43E-03	1.54
1719.7	6.46E-13	2.92E-13	3.77E-02	3.22E-02	5.83E-12	5.54E-12	1.96E-03	2.20E-03	1.54
1733.4	1.11E-12	5.00E-13	3.85E-02	3.29E-02	6.53E-12	6.19E-12	1.45E-03	1.62E-03	1.54

Table S4.
Polytherm

T (K)	Ga (Pa)	Ga _{corr} (Pa)	GaO (Pa)	Ga ₂ (Pa)	Ga ₂ O (Pa)	Ga ₂ O _{corr} (Pa)	In (Pa)	In _{corr} (Pa)	InO (Pa)	In ₂ (Pa)	In ₂ O (Pa)	In ₂ O _{corr} (Pa)	O _{2, meas} (Pa)	O _{2, calc} (Pa)
1538.8	1.20E-04	2.22E-05	1.51E-06	1.16E-05		1.10E-04	1.60E-03	4.20E-04	1.17E-06	1.03E-05	5.16E-05	1.24E-03	3.87E-03	6.95E-04
1547.2	1.24E-04	2.28E-05			8.03E-06	1.09E-04	1.62E-03	4.25E-04	1.33E-05	1.51E-05	5.30E-05	1.26E-03	4.66E-03	7.10E-04
1555.7	1.43E-04	2.65E-05	5.66E-05		8.12E-06	1.25E-04	1.76E-03	4.61E-04	2.23E-06	1.22E-05	4.46E-05	1.35E-03	4.52E-03	7.79E-04
1564.2	1.48E-04	2.74E-05	6.79E-06			1.21E-04	1.75E-03	4.61E-04	8.77E-06	6.74E-06	4.63E-05	1.35E-03	4.63E-03	7.76E-04
1577.9	2.40E-04	4.43E-05			3.73E-06	1.99E-04	2.07E-03	5.43E-04	4.39E-06	5.02E-06	3.84E-05	1.57E-03	5.04E-03	9.47E-04
1587.4	3.00E-04	5.54E-05				2.44E-04	2.13E-03	5.60E-04	1.57E-05	1.26E-05	4.74E-05	1.63E-03	5.70E-03	1.02E-03
1598.0	3.26E-04	6.02E-05			1.38E-05	2.80E-04	2.43E-03	6.39E-04	1.19E-05	7.26E-06	4.50E-05	1.84E-03	5.30E-03	1.16E-03
1609.7	4.20E-04	7.75E-05		7.11E-06	1.80E-06	3.51E-04	2.59E-03	6.81E-04	9.41E-06	4.68E-06	4.05E-05	1.96E-03	6.06E-03	1.29E-03
1618.1	4.92E-04	9.08E-05			8.82E-06	4.10E-04	2.87E-03	7.53E-04	7.79E-06	9.49E-06	3.35E-05	2.16E-03	5.55E-03	1.45E-03
1631.9	6.12E-04	1.13E-04	1.51E-05		1.31E-06	5.01E-04	2.94E-03	7.72E-04	5.86E-06	1.14E-05	3.42E-05	2.21E-03	5.49E-03	1.59E-03
1642.5	8.24E-04	1.52E-04												

Polytherm

T (K)	K (Ga)	SD	$\gamma(\text{GaO}_{1.5})$	SD	K (In)	SD	$\gamma(\text{InO}_{1.5})$	SD	AIW
1538.8	2.87E-15	1.18E-15	3.56E-01	2.97E-01	2.03E-13	1.14E-13	3.35E-02	2.75E-02	2.98
1547.2	3.00E-15	1.23E-15	2.50E-01	2.08E-01	2.09E-13	1.17E-13	2.50E-02	2.04E-02	2.89
1555.7	3.73E-15	1.53E-15	2.09E-01	1.74E-01	2.43E-13	1.36E-13	2.11E-02	1.73E-02	2.84
1564.2	3.84E-15	1.57E-15	1.46E-01	1.22E-01	2.42E-13	1.35E-13	1.54E-02	1.25E-02	2.74
1577.9	7.23E-15	2.93E-15	1.47E-01	1.22E-01	3.31E-13	1.84E-13	1.27E-02	1.03E-02	2.68
1587.4	9.53E-15	3.84E-15	1.27E-01	1.05E-01	3.60E-13	1.99E-13	9.78E-03	7.94E-03	2.60
1598.0	1.15E-14	4.59E-15	9.55E-02	7.92E-02	4.54E-13	2.50E-13	8.45E-03	6.85E-03	2.55
1609.7	1.59E-14	6.32E-15	8.00E-02	6.61E-02	5.23E-13	2.86E-13	6.46E-03	5.22E-03	2.47
1618.1	2.04E-14	8.04E-15	7.11E-02	5.87E-02	6.32E-13	3.44E-13	5.81E-03	4.69E-03	2.43
1631.9	2.72E-14	1.05E-14	5.29E-02	4.35E-02	6.93E-13	3.72E-13	3.98E-03	3.19E-03	2.33
1642.5									

3_high (KN_28_02_20_Ga_In_10000)

In-situ measurements of major- and minor elements.

1_low (KN_07-11-19_Ga-In-Tl_1000)

Table S5. EPMA – major element oxides (wt. %)															
Left Profile															
X-POS	Y-POS	Distance from interior (mm)													
			SiO ₂	1σ	Al ₂ O ₃	1σ	MgO	1σ	CaO	1σ	FeO	1σ	TiO ₂	1σ	Total
-28.92	32.78	0.000	50.41	0.11	15.34	0.06	10.36	0.06	23.70	0.07	0.064	0.009	0.000	0.000	99.87
-28.88	32.72	0.067	50.48	0.11	15.29	0.06	10.32	0.06	23.69	0.07	0.067	0.009	0.000	0.000	99.85
-28.84	32.66	0.138	50.51	0.11	15.27	0.06	10.45	0.06	23.54	0.07	0.067	0.009	0.000	0.000	99.85
-28.80	32.60	0.209	50.76	0.11	15.14	0.06	10.35	0.06	23.47	0.07	0.066	0.009	0.009	0.013	99.80
-28.74	32.53	0.308	50.80	0.11	15.20	0.06	10.50	0.06	23.29	0.07	0.057	0.009	0.009	0.013	99.86
-28.72	32.49	0.352	50.67	0.11	15.29	0.06	10.30	0.06	23.54	0.07	0.061	0.009	0.000	0.000	99.86
-28.67	32.42	0.433	50.66	0.11	15.26	0.06	10.41	0.06	23.44	0.07	0.068	0.009	0.005	0.014	99.85
-28.60	32.31	0.565	50.72	0.11	15.33	0.06	10.38	0.06	23.31	0.07	0.055	0.009	0.018	0.013	99.81
-28.56	32.25	0.637	50.71	0.11	15.27	0.06	10.43	0.06	23.38	0.07	0.080	0.009	0.011	0.013	99.88
-28.52	32.20	0.708	50.75	0.11	15.43	0.06	10.37	0.06	23.28	0.07	0.062	0.009	0.003	0.013	99.89
-28.47	32.14	0.784	50.81	0.11	15.43	0.06	10.37	0.06	23.19	0.07	0.059	0.009	0.000	0.000	99.87

Table S6.
LA-ICP-MS
Left Profile

X-POS	Y-POS	Distance between points (μm)	Distance from interior (μm)	Ga (ppm)	1σ	In (ppm)	1σ
11373	50275	0.0	0.0	434	6.1	0.051	0.010
11420	50243	56.9	56.9	426.3	6.2	0.041	0.007
11466	50212	55.5	112.3	427.9	5.5	0.040	0.007
11513	50180	56.9	169.2	425.2	5.5	0.038	0.009
11559	50149	55.5	224.7	424.9	5.6	0.035	0.006
11606	50118	56.3	281.0	421.6	6.5	0.032	0.006
11653	50086	56.9	337.8	419.5	5.9	0.032	0.008
11699	50055	55.5	393.3	428.4	5.9	0.030	0.007
11746	50023	56.9	450.2	429.9	6.6	0.032	0.012
11792	49992	55.5	505.6	426.2	6.5	0.023	0.007
11839	49961	56.3	561.9	418.8	6.1	0.015	0.005
11886	49929	56.9	618.8	424.3	6.9	0.014	0.006
11932	49898	55.5	674.3	424.9	5.4	0.009	0.005
11979	49866	56.9	731.1	415.6	5.2	0.012	0.005
12018	49840	46.9	778.0	413.8	6.4	0.012	0.006

Right Profile

17564	52347	0.0	0.0	437.5	5.8	0.031	0.008
17533	52292	63.1	63.1	439.1	5.5	0.029	0.008
17501	52237	63.6	126.8	432.1	7	0.025	0.008
17470	52183	62.3	189.0	442.5	5.3	0.023	0.007
17438	52128	63.6	252.7	432.5	5.7	0.029	0.008
17406	52073	63.6	316.3	437.1	6.2	0.025	0.008
17375	52018	63.1	379.4	436.7	5.9	0.020	0.006
17343	51963	63.6	443.1	436	4.9	0.019	0.006
17312	51909	62.3	505.3	433.6	7.1	0.020	0.006
17280	51854	63.6	569.0	435.5	6.7	0.018	0.005
17248	51799	63.6	632.6	430.6	6.3	0.013	0.005
17217	51744	63.1	695.7	436.5	6.1	0.014	0.005
17185	51689	63.6	759.4	433.5	5.5	0.011	0.005
17154	51635	62.3	821.6	429.5	6.4	0.007	0.004
17127	51589	53.3	875.0	425.5	7	0.013	0.006

2_low (KN_06_03_20_Ga_In_1000)

Table S7.
EPMA – major element oxides (wt. %)

Left Profile

X-POS	Y-POS	Distance from interior (mm)	SiO ₂	1σ	Al ₂ O ₃	1σ	MgO	1σ	CaO	1σ	Total
-25.22	32.26	0.000	50.29	0.11	15.53	0.06	10.84	0.06	23.17	0.07	99.84
-25.23	32.24	0.024	50.41	0.11	15.50	0.06	10.77	0.06	23.14	0.07	99.83
-25.25	32.22	0.048	50.16	0.11	15.55	0.06	10.86	0.06	23.27	0.07	99.84
-25.26	32.20	0.072	50.16	0.11	15.53	0.06	10.87	0.06	23.28	0.07	99.84
-25.27	32.18	0.096	50.24	0.11	15.53	0.06	10.85	0.06	23.22	0.07	99.84
-25.28	32.16	0.120	50.44	0.11	15.53	0.06	10.75	0.06	23.12	0.07	99.84
-25.29	32.14	0.144	50.40	0.11	15.48	0.06	10.79	0.06	23.16	0.07	99.83
-25.30	32.11	0.168	50.39	0.11	15.48	0.06	10.75	0.06	23.21	0.07	99.84
-25.31	32.09	0.192	50.43	0.11	15.50	0.06	10.73	0.06	23.18	0.07	99.84
-25.33	32.07	0.216	50.36	0.11	15.54	0.06	10.74	0.06	23.20	0.07	99.84
-25.34	32.05	0.240	50.61	0.11	15.49	0.06	10.60	0.06	23.12	0.07	99.82
-25.35	32.03	0.264	50.49	0.11	15.51	0.06	10.66	0.06	23.16	0.07	99.82
-25.36	32.01	0.289	50.38	0.11	15.51	0.06	10.72	0.06	23.21	0.07	99.82
-25.37	31.99	0.312	50.48	0.11	15.44	0.06	10.73	0.06	23.17	0.07	99.82
-25.38	31.97	0.336	50.37	0.11	15.50	0.06	10.72	0.06	23.22	0.07	99.81
-25.39	31.95	0.360	50.44	0.11	15.44	0.06	10.72	0.06	23.23	0.07	99.83
-25.41	31.92	0.384	50.52	0.11	15.49	0.06	10.68	0.06	23.16	0.07	99.85
-25.42	31.90	0.408	50.52	0.11	15.47	0.06	10.69	0.06	23.16	0.07	99.85
-25.43	31.88	0.432	50.52	0.11	15.46	0.06	10.65	0.06	23.20	0.07	99.84
-25.44	31.86	0.456	50.48	0.11	15.46	0.06	10.68	0.06	23.22	0.07	99.84
-25.45	31.84	0.480	50.57	0.11	15.46	0.06	10.61	0.06	23.21	0.07	99.85
-25.46	31.82	0.504	50.45	0.11	15.45	0.06	10.73	0.06	23.21	0.07	99.84
-25.47	31.80	0.528	50.46	0.11	15.46	0.06	10.73	0.06	23.21	0.07	99.85
-25.48	31.78	0.552	50.39	0.11	15.52	0.06	10.69	0.06	23.24	0.07	99.84
-25.50	31.76	0.576	50.41	0.11	15.47	0.06	10.72	0.06	23.25	0.07	99.85
-25.51	31.73	0.600	50.45	0.11	15.46	0.06	10.67	0.06	23.26	0.07	99.84
-25.52	31.71	0.624	50.45	0.11	15.48	0.06	10.69	0.06	23.21	0.07	99.83
-25.53	31.69	0.648	50.52	0.11	15.47	0.06	10.63	0.06	23.21	0.07	99.83
-25.54	31.67	0.672	50.47	0.11	15.38	0.06	10.63	0.06	23.35	0.07	99.83
-25.55	31.65	0.696	50.43	0.11	15.46	0.06	10.69	0.06	23.25	0.07	99.83
-25.56	31.63	0.720	50.40	0.11	15.46	0.06	10.68	0.06	23.31	0.07	99.85
-25.58	31.61	0.744	50.46	0.11	15.48	0.06	10.65	0.06	23.26	0.07	99.85
-25.59	31.59	0.768	50.39	0.11	15.46	0.06	10.65	0.06	23.32	0.07	99.83
-25.60	31.57	0.792	50.35	0.11	15.45	0.06	10.66	0.06	23.39	0.07	99.85
-25.63	31.50	0.864	50.30	0.11	15.47	0.06	10.68	0.06	23.40	0.07	99.85
-25.64	31.48	0.888	50.42	0.11	15.41	0.06	10.65	0.06	23.35	0.07	99.83
-25.66	31.46	0.912	50.38	0.11	15.40	0.06	10.67	0.06	23.38	0.07	99.83
-25.67	31.42	0.952	50.34	0.11	15.41	0.06	10.64	0.06	23.44	0.07	99.83

Right Profile

-30.17	32.22	0.000	50.32	0.11	15.45	0.06	10.73	0.06	23.33	0.07	99.83
-30.16	32.19	0.029	50.39	0.11	15.48	0.06	10.66	0.06	23.30	0.07	99.84
-30.15	32.17	0.059	50.44	0.11	15.47	0.06	10.69	0.06	23.23	0.07	99.83
-30.13	32.14	0.089	50.44	0.11	15.47	0.06	10.71	0.06	23.21	0.07	99.82
-30.12	32.11	0.118	50.49	0.11	15.49	0.06	10.66	0.06	23.19	0.07	99.83

-30.09	32.03	0.206	50.34	0.11	15.51	0.06	10.74	0.06	23.25	0.07	99.84
-30.08	32.00	0.236	50.27	0.11	15.51	0.06	10.76	0.06	23.29	0.07	99.83
-30.07	31.98	0.265	50.48	0.11	15.49	0.06	10.66	0.06	23.21	0.07	99.84
-30.06	31.95	0.294	50.53	0.11	15.48	0.06	10.65	0.06	23.19	0.07	99.85
-30.04	31.92	0.324	50.55	0.11	15.47	0.06	10.68	0.06	23.13	0.07	99.84
-30.01	31.84	0.413	50.53	0.11	15.42	0.06	10.70	0.06	23.16	0.07	99.81
-30.00	31.81	0.442	50.50	0.11	15.43	0.06	10.67	0.06	23.22	0.07	99.83
-29.99	31.79	0.471	50.45	0.11	15.48	0.06	10.71	0.06	23.19	0.07	99.83
-29.98	31.76	0.501	50.49	0.11	15.43	0.06	10.74	0.06	23.16	0.07	99.83
-29.97	31.73	0.531	50.46	0.11	15.49	0.06	10.72	0.06	23.16	0.07	99.83
-29.95	31.70	0.560	50.46	0.11	15.52	0.06	10.70	0.06	23.14	0.07	99.82
-29.94	31.68	0.589	50.48	0.11	15.45	0.06	10.74	0.06	23.17	0.07	99.84
-29.93	31.65	0.619	50.48	0.11	15.49	0.06	10.69	0.06	23.19	0.07	99.85
-29.92	31.62	0.648	50.48	0.11	15.50	0.06	10.68	0.06	23.18	0.07	99.84
-29.91	31.60	0.678	50.45	0.11	15.53	0.06	10.66	0.06	23.19	0.07	99.84
-29.90	31.57	0.707	50.39	0.11	15.53	0.06	10.69	0.06	23.22	0.07	99.83

Table S8.
LA-ICP-MS

Left Profile

X-POS	Y-POS	Distance from interior (μm)	Ga (ppm)	1σ	In (ppm)	1σ
47040	17467	0.0	668.6	7.4	0.80	0.02
47062	17387	83.0	667.2	7.8	0.69	0.02
47085	17314	159.5	667.5	9.1	0.69	0.03
47114	17232	246.5	662.6	9.3	0.52	0.02
47144	17149	334.7	675.9	8.5	0.49	0.02
47168	17078	409.7	678.8	7.3	0.43	0.02
47198	16992	500.8	681.8	7.2	0.41	0.01
47219	16916	579.6	678.3	7.0	0.40	0.01
47245	16838	661.8	679.2	6.9	0.49	0.04
47271	16753	750.7	684.7	7.2	0.40	0.01
47294	16671	835.9	681.9	7.1	0.33	0.01
47316	16597	913.1	715.3	5.1	0.14	0.01

Right Profile

51874	16715	0.0	726.2	6.6	5.55	0.07
51843	16651	71.1	720.2	6.7	5.66	0.25
51816	16597	131.5	719.0	7.1	5.10	0.07
51781	16545	194.2	722.4	6.1		
51748	16485	262.6	710.5	9.9	4.57	0.06
51722	16432	321.7	703.8	8.8	4.12	0.08
51679	16371	396.3	700.9	9.4	3.65	0.06
51641	16320	459.9	708.0	9.7	3.04	0.05
51602	16264	528.2	706.9	7.9	2.38	0.04
51559	16206	600.4	697.5	9.2	1.71	0.03
51531	16147	665.7	699.2	9.3	1.08	0.02
51506	16110	710.3	700.2	7.7	0.66	0.02
51479	16072	756.9	735.1	6.5	0.19	0.02

3_high (KN_28_02_20_Ga_In_10000)

Table S9.
EPMA - major element oxides (wt. %)

X-POS	Y-POS	Distance from interior (mm)	SiO ₂	1σ	Al ₂ O ₃	1σ	MgO	1σ	CaO	1σ	Total
29.77	33.72	0.000	49.80	0.11	15.30	0.06	10.68	0.06	22.57	0.07	98.36
29.75	33.69	0.034	49.90	0.11	15.24	0.06	10.63	0.06	22.59	0.07	98.36
29.73	33.67	0.069	49.91	0.11	15.24	0.06	10.62	0.06	22.60	0.07	98.37
29.71	33.64	0.103	49.86	0.11	15.22	0.06	10.59	0.06	22.69	0.07	98.35
29.69	33.61	0.137	49.98	0.11	15.24	0.06	10.53	0.06	22.64	0.07	98.39
29.65	33.56	0.206	50.02	0.11	15.26	0.06	10.50	0.06	22.58	0.07	98.35
29.63	33.53	0.240	49.72	0.11	15.21	0.06	10.67	0.06	22.76	0.07	98.37
29.60	33.50	0.274	49.79	0.11	15.20	0.06	10.63	0.06	22.75	0.07	98.38
29.58	33.47	0.309	49.77	0.11	15.17	0.06	10.66	0.06	22.79	0.07	98.39
29.56	33.45	0.343	49.83	0.11	15.12	0.06	10.62	0.06	22.82	0.07	98.39
29.54	33.42	0.377	49.81	0.11	15.13	0.06	10.62	0.06	22.85	0.07	98.41
29.52	33.39	0.412	49.77	0.11	15.10	0.06	10.66	0.06	22.88	0.07	98.40
29.50	33.36	0.446	49.70	0.11	15.14	0.06	10.63	0.06	22.92	0.07	98.39
29.48	33.34	0.480	49.65	0.11	15.14	0.06	10.63	0.06	22.99	0.07	98.41
29.46	33.31	0.515	49.86	0.11	15.14	0.06	10.55	0.06	22.87	0.07	98.41
29.37	33.20	0.653	49.65	0.11	15.11	0.06	10.67	0.06	22.96	0.07	98.40
29.35	33.18	0.686	49.70	0.11	15.12	0.06	10.59	0.06	23.00	0.07	98.40

Table S10.
LA-ICP-MS

X-POS	Y-POS	Distance from interior (μm)	Ga (ppm)	1σ	In (ppm)	1σ
114690	16523	0.0	7979	64	1715	12
114634	16487	66.6	7909	54	1627	12
114577	16459	130.1	7899	64	1508	11
114517	16410	207.5	7919	70	1333	11
114456	16374	278.4	7923	62	1160	11
114385	16329	362.4	7960	71	1017	9
114249	16193	554.8	8036	126	890	7
114191	16154	624.7	8066	68	913	7
114132	16118	693.8	8075	64	985	7

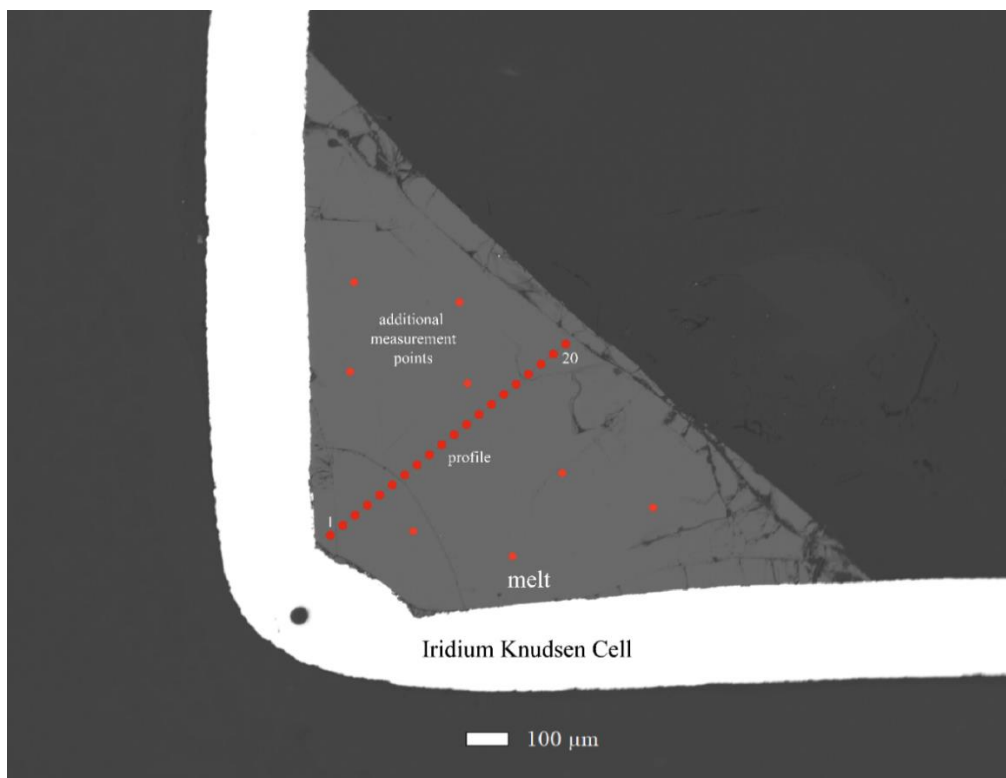


Fig. S1: Measurement points from EPMA for run 3_high.

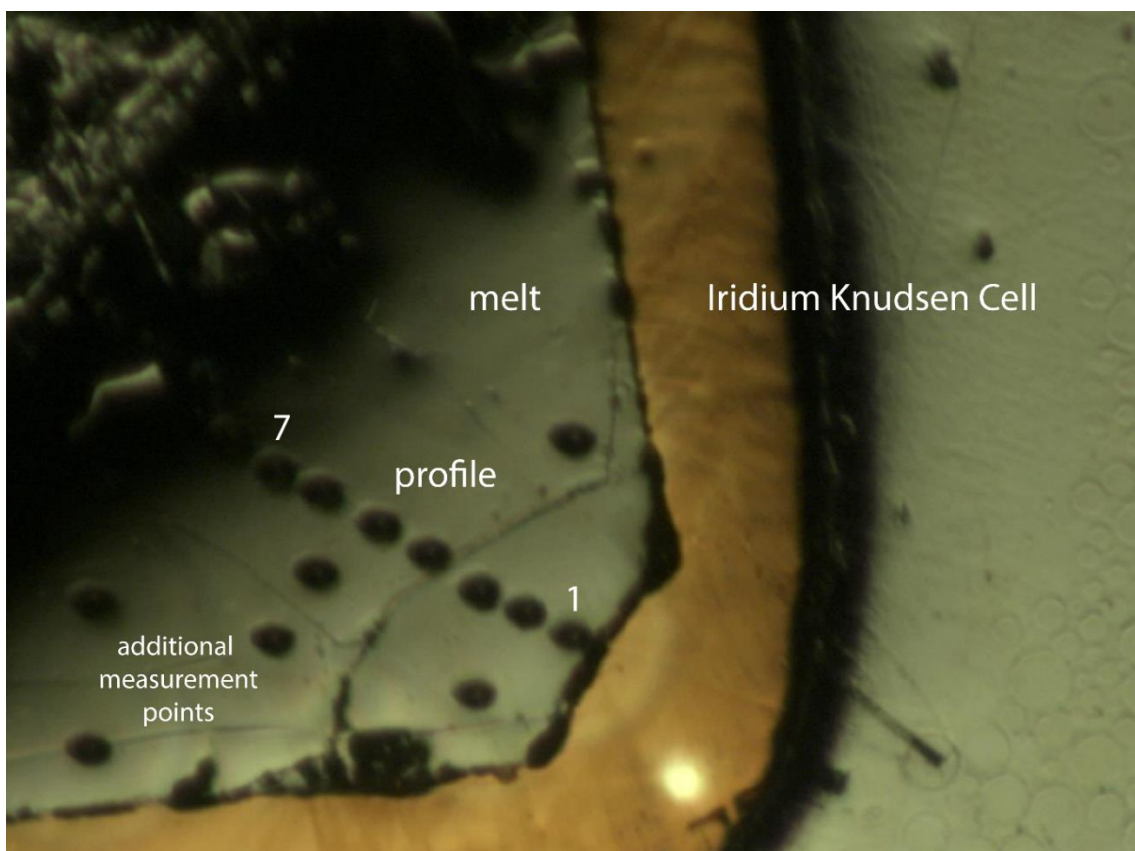


Fig. S2: Measurement points with the LA-ICP-MS.

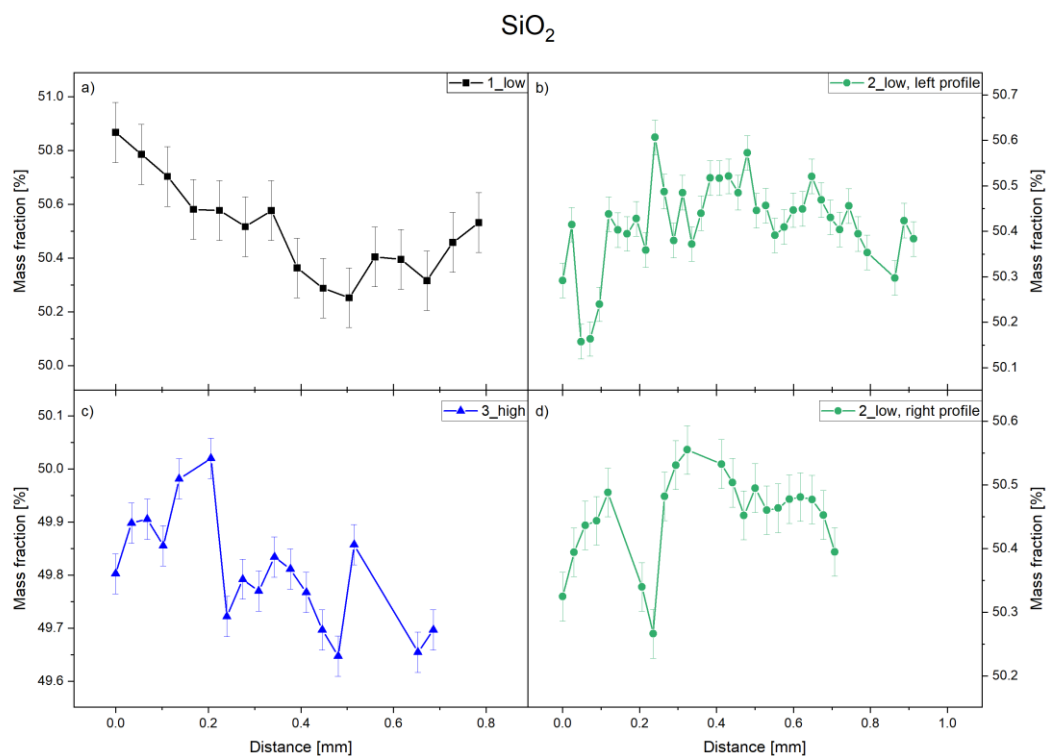


Fig. S3: Results from EPMA profile for SiO_2 . Distance is measured from measurement point 1 in the inner part of the melt to the last measurement point at the surface (*cf.* Fig S1).

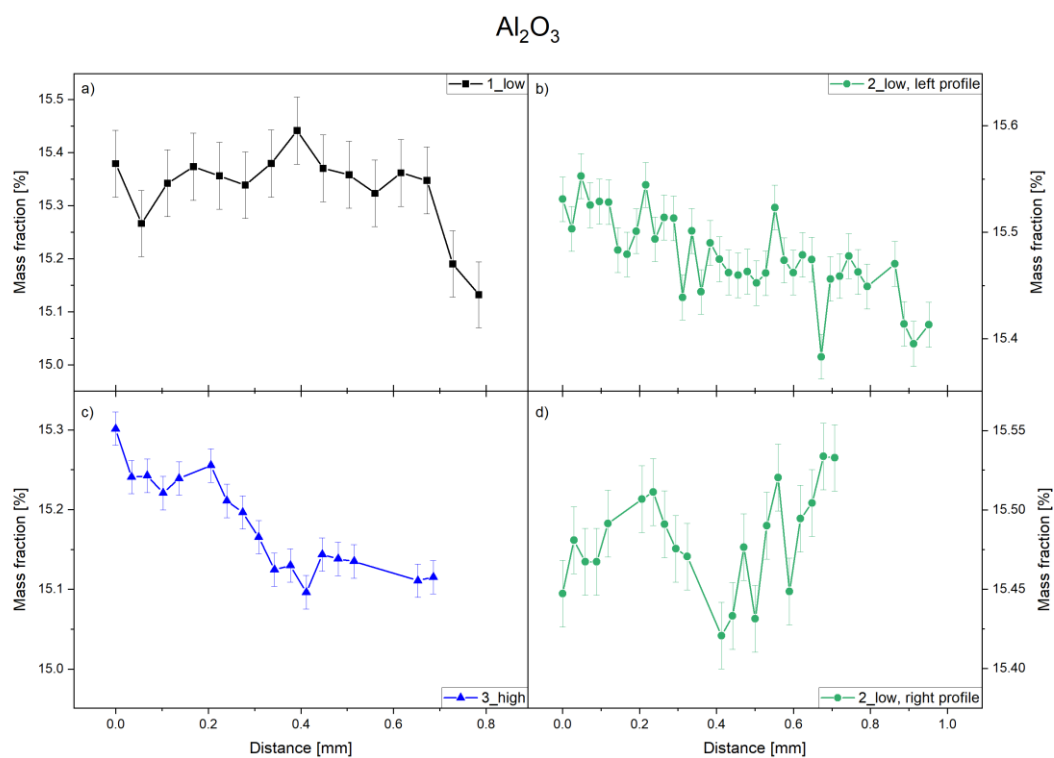


Fig. S4: Results from EPMA profile for Al_2O_3 . Distance is measured from measurement point 1 in the inner part of the melt to the last measurement point at the surface (*cf.* Fig. S1).

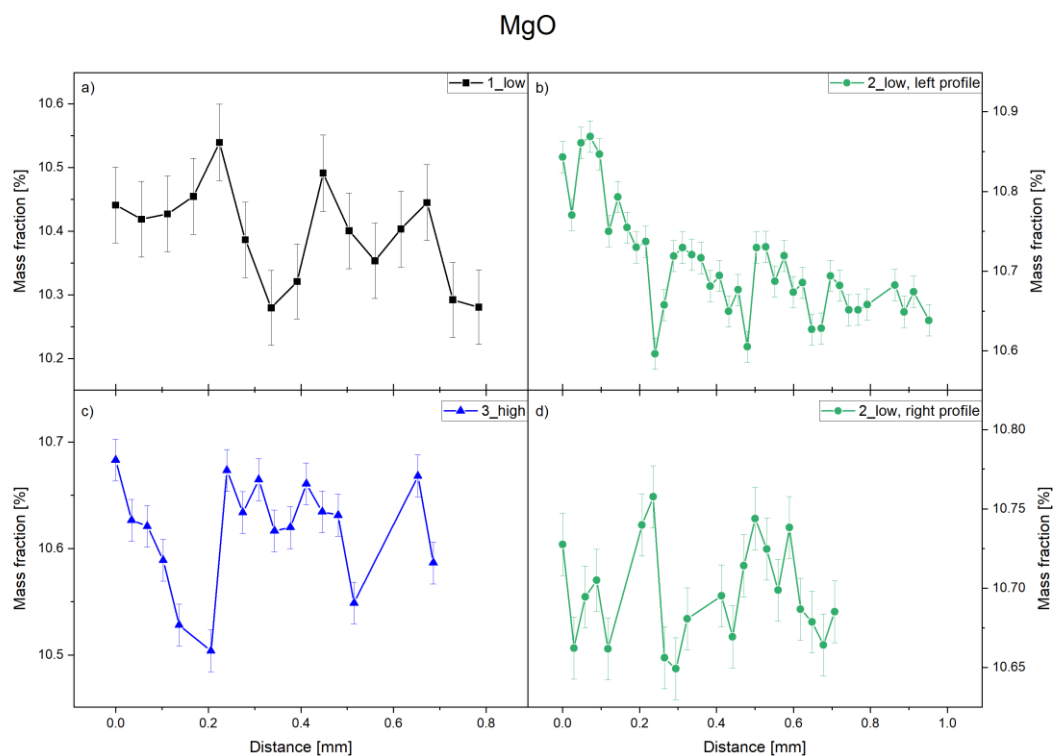


Fig. S5: Results from EPMA profile for MgO. Distance is measured from measurement point 1 in the inner part of the melt to the last measurement point at the surface (*cf.* Fig. S1).

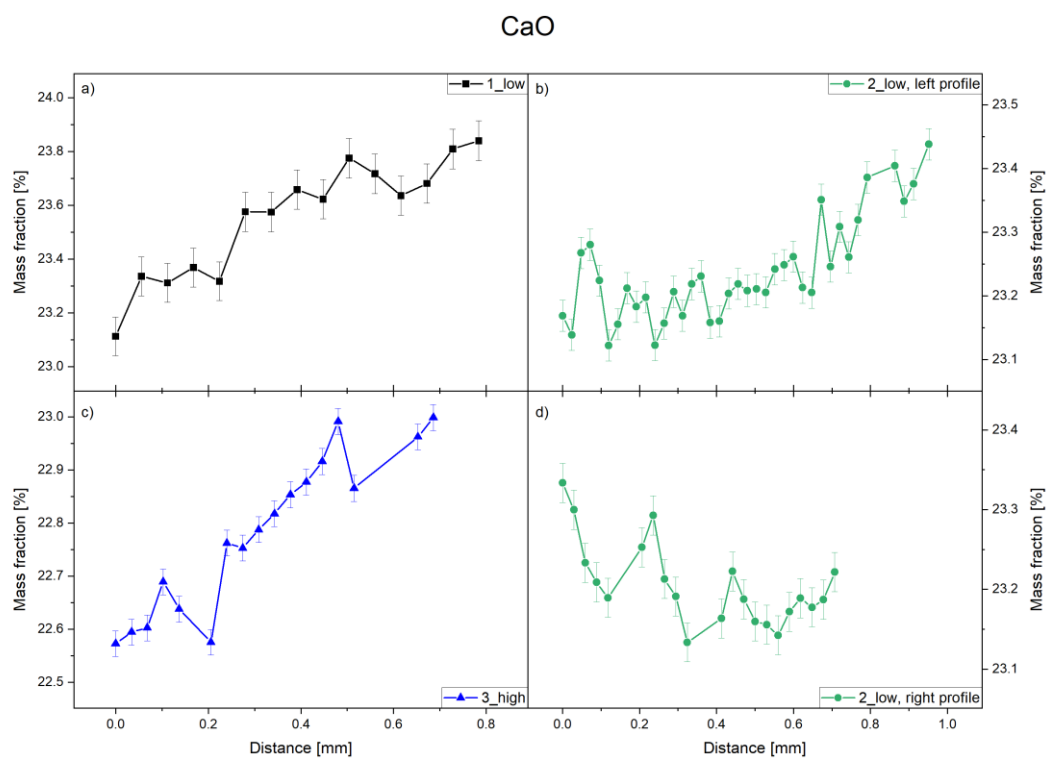


Fig. S6: Results from EPMA profile for CaO. Distance is measured from measurement point 1 in the inner part of the melt to the last measurement point at the surface (*cf.* Fig. S1).

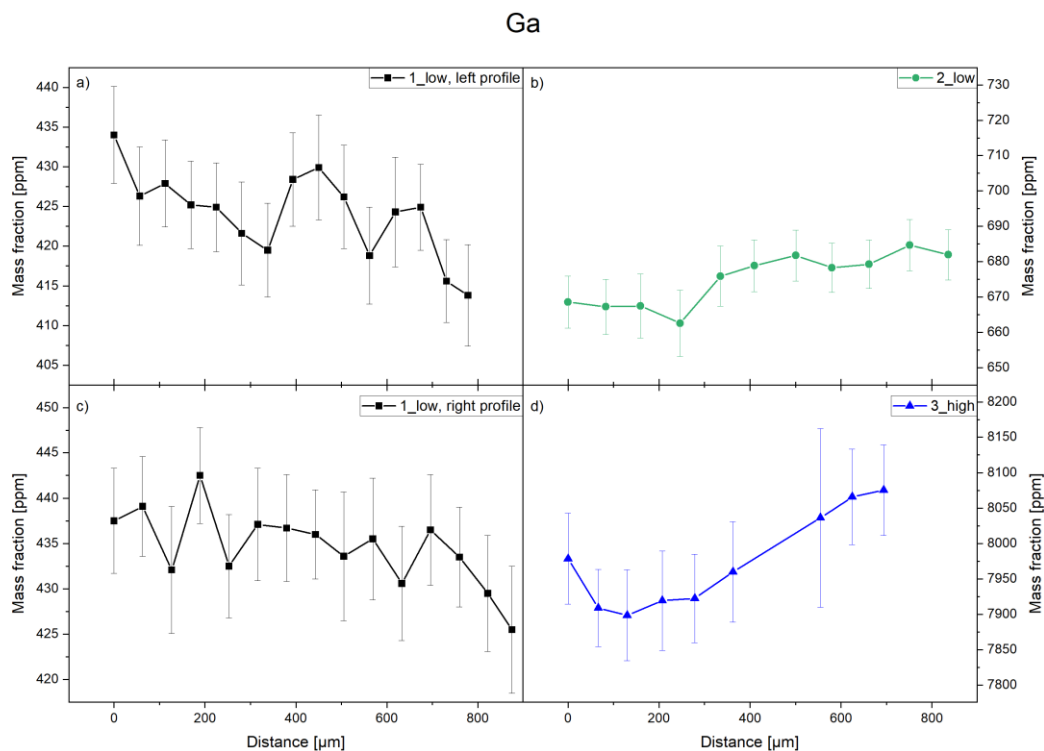


Fig. S7: Results from LA-ICP-MS profile for Ga. Distance is measured from measurement point 1 in the inner part of the melt to the last measurement point at the surface (*cf.* Fig. S2).

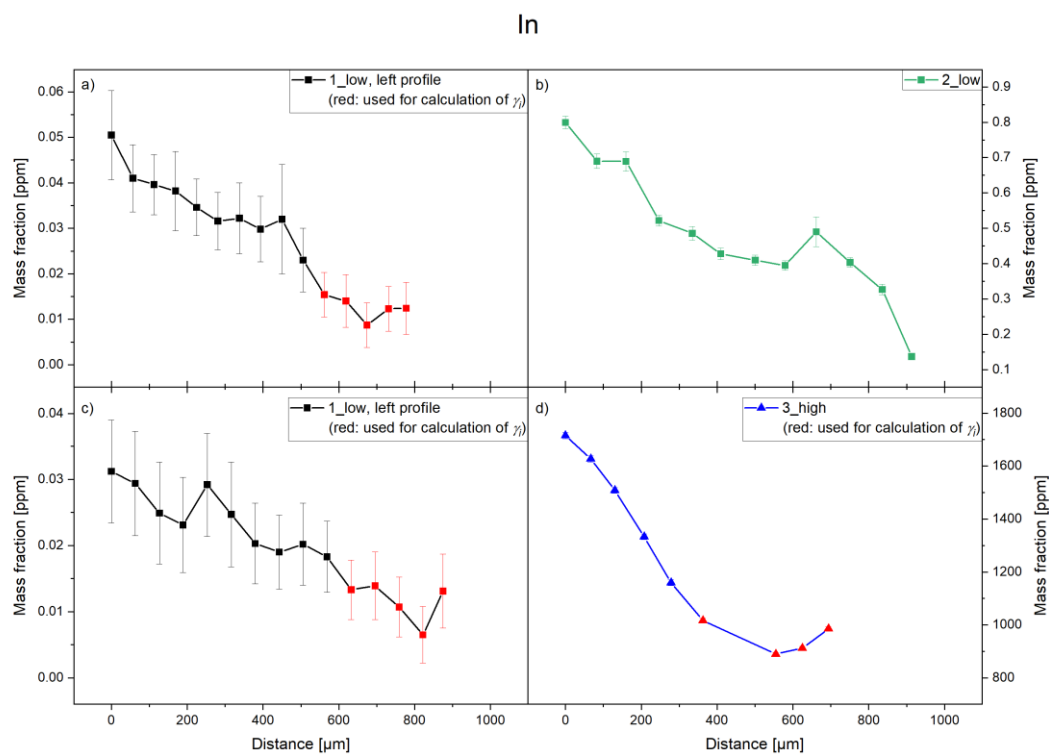


Fig. S8: Results from LA-ICP-MS profile for In. Distance is measured from measurement point 1 in the inner part of the melt to the last measurement point at the surface (*cf.* Fig. S2).

Calculation of volatilities of trace elements.

The volatilities of Ga and K or In and Zn, respectively, are compared on the basis of the magnitude of the partial pressures of the respective elements at a given temperature oxygen fugacity fO_2 at the iron-wüstite buffer IW (eq. S1).

$$p(M) = \frac{K_{p.o.} a(MO_m) p^0}{\left(\frac{p(O_2)}{p^0}\right)^n} \quad (S1)$$

Here, m and n are the stoichiometric coefficients of the particular vaporisation reaction. The equilibrium constants for the vaporisation reactions above the pure oxides, $K_{p.o.}$, for Ga and In are calculated as described before in eq. 20 and 21 (Bischof *et al.*, 2023), for K (eq. S2)

$$K = \exp\left(-\frac{209500 \frac{J}{mol} - T * 50 \frac{J}{mol K}}{RT}\right) \quad (S2)$$

and Zn (eq. C3)

$$K = \exp\left(-\frac{548600 \frac{J}{mol} - T * 253 \frac{J}{mol K}}{RT}\right) \quad (S3)$$

according to Sossi *et al.* (2019). Activities $a(MO_m)$ are calculated with eq. 20, the mole fraction x_i is set to 0.0001 for all of the four elements. For Ga and In, the activity coefficients γ_i , determined within this work are applied, for K and Zn it is set to 1, since it is already implied in the values for ΔH and ΔS in eq. S2 and S3. The oxygen fugacity around the iron-wüstite buffer is calculated using eq. S4 (O'Neill and Pownceby, 1993),

$$fO_{2\Delta IW X} = 10^{\left(\frac{\Delta G_{Fe-FeO,T}}{RT \ln(10)}\right) \pm X} \quad (C4)$$

where X is the difference from the iron-wüstite buffer and $\Delta G_{Fe-FeO,T}$ the Gibbs free energy of the buffer reaction (O'Neill and Pownceby, 1993).